Interpreting A/B test results: false negatives and power

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This is the fourth post in a multi-part series on how Netflix uses A/B tests to inform decisions and continuously innovate on our products. Need to catch up? Have a look at Part 1 (Decision Making at Netflix), Part 2 (What is an A/B Test?), Part 3 (False positives and statistical significance). Subsequent posts will go into more details on experimentation across Netflix, how Netflix has invested in infrastructure to support and scale experimentation, and the importance of the culture of experimentation within Netflix.

In *Part 3: False positives and statistical significance*, we defined the two types of mistakes that can occur when interpreting test results: false positives and false negatives. We then used simple thought exercises based on flipping coins to build intuition around false positives and related concepts such as statistical significance, p-values, and confidence intervals. In this post, we'll do the same for false negatives and the related concept of statistical power.

Figure 1: As in Part 3, we'll use thought exercises based on flipping coins, such as this one displaying Caesar Augustus, to build up intuition about core statistical concepts.

False negatives and power

A **false negative** occurs when the data do not indicate a meaningful difference between treatment and control, but in truth there is a difference. Continuing on an example from Part 3, a false negative corresponds to labeling the photo of the cat as a “not cat.” False negatives are closely related to the statistical concept of power, which gives the probability of a true positive *given the experimental design and a true effect of a specific size*. In fact, power is simply one minus the false negative rate.

Power involves thinking about possible outcomes given a specific assumption about the actual state of the world — similar to how in Part 3 we defined significance by first assuming the null hypothesis is true. To build intuition about power, let’s go back to the same coin example from Part 3, where the goal is to decide if the coin is unfair using an experiment that calculates the fraction of heads in 100 flips. The distribution of outcomes under the null hypothesis that the coin is fair is shown in black in Figure 2. To make the diagram easier to interpret, we’ve smoothed over the tops of the histograms.

What would happen in this experiment if the coin is ***not*** fair? To make the thought exercise more specific, let’s work through what happens when we have a coin where heads occurs, on average, 64% of the time (the choice of that peculiar number will become clear later on). Because there is uncertainty or noise in our experiment, we don’t expect to see exactly 64 heads in 100 flips. But as with the null hypothesis that the coin is fair, we can calculate all the possible outcomes if this specific alternative hypothesis is true. This distribution is shown with the red curve in Figure 2.

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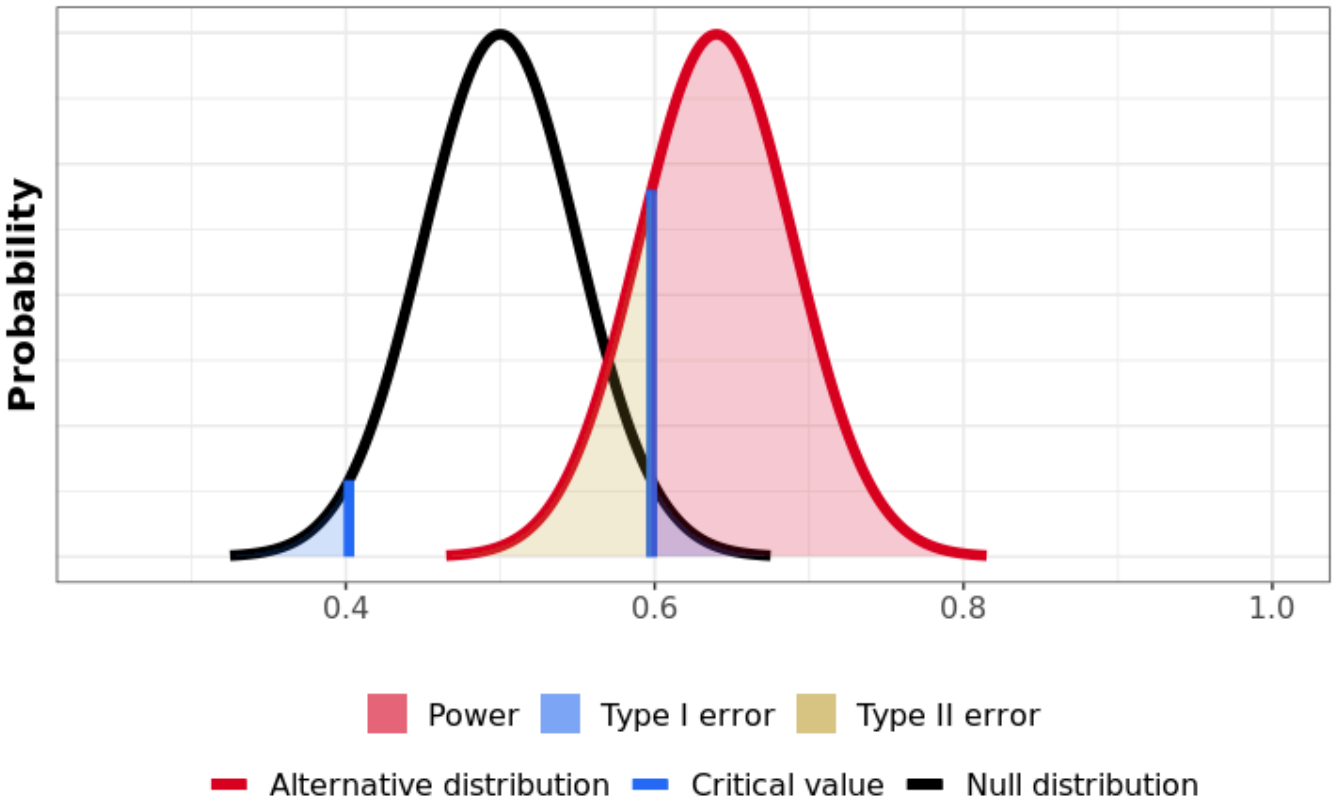

Figure 2: Illustrating power using the example of flipping a coin 100 times and calculating the fraction of heads. The black and red dashed lines show, respectively, the distribution of outcomes assuming the probability of heads is 50% (null hypothesis) and 64% (specific value of the alternative hypothesis). Here, the power against this alternative is 80% (red shading).

Visually, power is the fraction of this alternative (red) distribution that lies beyond the critical values under the null hypothesis (the blue lines and black curve; see Part 3). Here, 80% of the alternative distribution (red) falls to the right of the taller blue line that demarcates the critical value of the upper rejection region. Assuming that the truth about the coin is that the probability of heads is 64%, then the power of this test is 80%. To be complete, there is also a negligibly small part of the alternative (red) distribution that falls within the lower rejection region (to the left of the short blue line).

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The power of a test corresponds to a specific, postulated effect size. In our example, the test has 80% power to detect that a coin is unfair, if that unfair coin in truth has a probability of heads equal to 64%. The interpretation is as follows: if the coin has probability of heads equal to 64%, and we repeatedly run the experiment of flipping 100 times and making a decision at the 5% significance level, then we will correctly reject the null hypothesis that the coin is fair in about 4 out of every 5 experiments. And 20% of those repeated experiments will result in a false negative: we'll not reject the null hypothesis that the coin is fair, even though it is unfair.

Ways to increase power

In designing an A/B test, we first fix the significance level (the convention is 5%: if there is no difference between treatment and control, we'll see false positives 5% of the time), and then design the experiment to control false negatives. There are three primary levers we can pull to increase power and reduce the probability of false negatives:

1. Simply put, the larger the effect size — the difference in metric values between Groups A and B — the higher the probability that we'll be able to correctly detect that difference. To build intuition, think about running an experiment to determine if a coin is unfair, where the data we collect is the fraction of heads in 100 flips. Now think of two scenarios. In the first scenario, the true probability of heads is 55%, and in the second it is 75%. Intuitively (and mathematically!) it is more likely that our experiment identifies the coin as unfair in the second scenario. The true probability of heads is further from the null value of 50%, so it's more likely that an experiment will produce an outcome that falls in the rejection region. In the product development context, we can increase the expected magnitude of metric movements by being bold vs incremental with the hypotheses we test. Another strategy to increase effect sizes is to test in new areas of the product, where there may be room for larger improvements in member satisfaction. That said, one of the joys of learning through experimentation is the element of surprise: at times, seemingly small changes can have a major impact on top-line metrics.
2. . The more units in the experiment, the higher the power and the easier it is to correctly identify smaller effects. To build intuition, think again about running an experiment to determine if a coin is unfair, where the data we collect is the fraction of heads in a fixed number of flips and the true probability of heads is 64%. Consider two scenarios: in the first, we flip the coin 20 times, and in the second, we flip the coin 100 times. Intuitively (and mathematically!), it is more likely that our experiment identifies the coin as unfair in the second scenario. With more data, the result from the experiment is going to be closer to the true rate of 64% heads, while the outcomes under the assumption of a fair coin concentrate around 0.50, causing the rejection region to encroach on the 50% value. These effects combine, so that with more data there is a greater probability that the result from the experiment with the unfair coin will fall in that rejection region, resulting in a true positive. In the product development context, we can increase the power by allocating more members (or other units) to the test or by reducing the number of test groups, though there is a tradeoff between the sample size in each test and the number of non-overlapping tests that can be run at the same time.

3. . The more homogenous the metric within the population we are testing on, the easier it is to correctly identify true effects. The intuition for this one is a bit trickier, and our simple coin examples finally break down. Say at Netflix that we run a test that aims to reduce some measure of latency, such as the delay between a member pressing play and video playback commencing. Given the variety of devices and internet connections that people use to access Netflix, there is a lot of natural variability in this metric across our users. As a result, if the test treatment results in a small reduction in the latency metric, it's hard to successfully identify — the "noise" from the variability across members overwhelms the small signal. In contrast, if we ran the test on a set of members that used similar devices with similar web connections, then the small signal is easier to identify — there is less noise that might drown out the signal. We spend a lot of time at Netflix building statistical analysis models that exploit this intuition, and increase power by effectively lowering the variability; see here for a technical description of our approach.

Powering for reasonable and meaningful effects

Power and the false negative rate are functions of a postulated effect size. Much like how the 5% false positive rate is a widely-accepted convention, the rule of thumb with power is to aim for 80% power for a **reasonable** and **meaningful** effect size (we'll get to each of those below). That is, we postulate an effect size and then design the experiment, primarily through setting the sample size, such that, if the true impact of the treatment experience is as we've postulated, the test will correctly identify that there is an effect 80% of the time. And 20% of the time the result from the test will be a false negative: in truth, there is an effect, but our observation from the test does not lie in the rejection region and we fail to conclude that there is an effect. That's why the examples above used a 64% probability of heads: an experiment with 100 flips then has 80% power.

What constitutes a **reasonable** effect size can be tricky, as tests can surprise us. But a mix of domain knowledge and common sense can generally provide solid estimates. In an area where testing has a long history, such as optimizing the recommendation systems that help Netflix members choose content that's great for them, we have a solid idea about the effect sizes that our tests tend to produce (be they positive or negative). Given an understanding of past effect sizes, as well as the analysis strategy, we can set the sample size to ensure the test has 80% power for a reasonable metric movement.

The second consideration, both in this experimental design phase and in deciding where to invest efforts, is to determine what constitutes a **meaningful** impact to the primary metrics used to decide the test. What is meaningful will depend on the impact area of the experiment (member satisfaction, playback latency, technical performance of back end systems, etc.), and potentially the effort or costs associated with the new product experience. As a hypothetical, say that, for effect sizes smaller than a 0.1% change in the primary metric, the cost of supporting the new product feature outweighs the benefits. In this case, there's little point in powering a test to detect a 0.01% change in the metric, as successfully identifying an effect of that size won't result in a meaningful change in decisions. Likewise, if the effect sizes seen in

tests in a given innovation area are consistently immaterial to the user experience or the business, that's a sign that experimentation resources can be more efficiently deployed elsewhere.

Summary

Parts 3 and 4 of this series have focussed on defining and building intuition around the core concepts used to analyze test results: false positives and negatives, statistical significance, p-values, and power.

An uncomfortable truth about experimentation is that we can't simultaneously minimize both false positives and false negatives. In fact, false positives and negatives trade off with one another. If we used a more stringent false positive rate, such as 0.01%, we'd reduce the number of false positives for tests where there is no difference between A and B — but we'd also reduce the power of the test, increasing the rate of false negatives, for those tests where there is a meaningful difference. Using a 5% false positive rate and targeting 80% power are well-established conventions that balance between limiting false discovery and enabling true discovery. However, in instances where a false positive (or false negative) poses a larger risk, researchers may deviate from these rules of thumb to minimize one type of uncertainty over another.

Our goal is not to eliminate uncertainty, but to understand and quantify the uncertainty in order to make sound decisions. In many cases, results from A/B tests require nuanced interpretation, and in fact the test result itself is only one input into a business decision. In the next post, we'll cover how to build confidence in a decision using test results. Follow the Netflix Tech Blog to stay up to date.